A Machine Learning Framework for Detecting Refund Abuse in E-Commerce Websites

Prof. Sumeet Shingi[1], Vedant Gandre[2], Devansh Mane[3] and Harshit Jain[4]

1,2,3,4*Computer Science Engineering (AIML), A.P. Shah Institute of Technology, Thane, 400615, Maharashtra, India.
Contributing authors: svshingi@apsit.edu.in, vedantgandre11@apsit.edu.in, devanshmane61@apsit.edu.in, harshitjain39@apsit.edu.in
†These authors contributed equally to this work.

Abstract. In online shopping Refund Abuse has turned into a real issue. Due to which it is leading to financial detriment across the e-commerce system. These fraudulent activities, which are systematically planned elude detection the traditional security setups the companies possess. This paper aims to introduce Fraud Lens, which is primarily a machine learning framework to detect fraudulent return patterns at an early stage. It attempts to identify these patterns, while still maintaining a positive user experience for regular users. Based on recent studies on behavioral patterns of people in e-commerce, integrated assistant tools is suitable in this scenario. The framework operates through a multi-stage pipeline that investigates the large-scale transaction data. The crucial component of this Fraud lens is the dual-purpose engine. At the first stage, a predictive model is employed that determines a specific Return Rate Percentage for every user, enabling the identification of high-risk users from legitimate customers. In the second stage, a new subsystem is introduced for comparing prices across platforms. This valuable functional unit transforms the system into something useful like a shopping assistant. The customers are provided with real-time market deals, which adds value. Simultaneously, it provides a baseline for spotting price-sensitive patterns in shopping behavior. Which in turn contributes to more robust fraud identification. These evaluated data are used by different ML models to compare and narratives about fraud are generated automatically using natural language processing. This helps users minimize the manual investigation efforts and mitigate operational latency.

Keywords: Machine Learning, Refund Abuse, E-Commerce, Return Rate Percentage, Price Comparison, Fraud Detection, Behavioral Analysis.

1 Introduction

In recent years, e-commerce has experienced rapid growth, it is seen that people are adopting online purchasing. Billions of users worldwide are dependent on digital transactions and so it has become a crucial mode of commerce. Moreover, online fraud has also seen a huge frequency of criminal acts. Based on previous research, observations focus on third- party aspects like unauthorized acquisition of user data. Despite this,

one major challenge still impacts business profitability is refund abuse, which is commonly referred to as first-party fraud.

Another significant thing seen in the market is when the consumers take advantage of these easy return rules. Such as purchasing clothes for temporary use and then returning them immediately, this type of behaviour is also known as wardrobing or another way is claiming that the package was never delivered so they can get a refund and also keep the item. As stated in [2], it is very evident that returns are no longer about logistics, it is also a crucial aspect of security, because it can hide larger patterns of exploitation.

Typical fraud systems miss this because everything looks normal when the user purchases anything. As pointed out by [10], these traditional rule-based setups process everything the same, but fraudsters keep changing how they do things. Due to this, we really need more advanced rules to distinguish between loyal users and fraudsters.

This paper introduces Fraud Lens; which is based on machine learning framework to get a system-wide view id customers are genuine. There are various such models but the problem is they just check for last recent order [3] [6], but fraud lens checks for the whole user history.

This Fraud Lens has two main units. Firstly, risk quantification computes a return rate percentage that automatically gets updated when the user purchases anything or returns the item, which in turn gives the admin a score to decide if an account needs to be supervised or blocked. Secondly, the price-comparison tool makes it user-friendly similar to the E-vigilant model in [11], which helps users by giving real-time deals to help people minimize expenditure in addition to this, it also monitors if low prices are linked to more returns, which reveals the behavioural patterns.

The main concept here is to not just spot fraud after it happens but to go beyond it to something more active with analysis. Contributions include finding weird return habits in big data sets by building a machine learning setup. Also, there is a price feature for keeping users around and watching behaviour. Additionally, Natural Language Processing is used to generate narratives that are interpretable from model outputs, so that it becomes easier for the investigator to understand.

Fraud Lens shows platforms can have enhanced security without making users agitated, by smart tech achieving this balance remains challenging but it might work even if it has potential concerns regarding simplification.

2 Literature Review

Despite having easy and reliable access to online ecommerce stores, certain weakness arises in their system at a rapid pace. Studies proves that security features have outgrown the login methodologies ages ago with now turning the attention towards the behavioral patterns hidden in the customer data. In replacement of passwords, complex algorithms that detect irregularities by analyzing the purchase history by the customer. These automated and smart tools handle large volume of data provided by the platform efficiently on a daily basis in a way which will be impossible for a team of humans to review by hand. Tough choices emerge when the convenience is met with the risk that might be hidden deep in the surface of dataset.

In real life scenarios, A movement in the field follows a pattern labelled "Multi-stage pipelines" by specialist [3]. Instead of going by each transaction separately, these systems search for repeating behaviors data. By providing a large amount of data, these algorithms distinguish errors from clear manipulations. In this procedure accuracy matters deeply; research provides proof that overly strict controls may label an activity as fraud where it may be [5][13]. In this crucial system success means more than just being correct- it involves clarity, an ability to offer clear data representation for human reviewer [6].

Another pattern shapes how systems are engaging different user interests. Certain applications don't only defend some organizational needs, but they also support individual needs. These designs carry the label of "E-Vigilant"[11]. Where some data analyst applies certain means such as "image based" methods to reduce unintentional returned products [4], some other studies provide solid proof behind the pattern of these deliberate product returns [2]. This is the point where financial losses are accumulated silently- which is commonly referred as "The hidden bleed" in online retail. An absence exists in research of systems that are capable of finding habitual returners prior to the loss incurrence.

Still transparency is faded when machines gain influence that raises concern about hidden decision making. Due to the clarity problems, researchers now seek to gain deep insights into systems that reveal their reasonings. A fraud lent tracker system might clarify risks by generating simple reports which can be sent to the respective staffs to further clarify the activities [7]. Such openness support fairness, meanwhile also reducing chances of software favoring one group over the other unknowingly.

Lately, there are efforts taken to bring consistency to risk measurement methods. Increasingly, further advancements in the field of Artificial Intelligence developed Graph Neural Networks to uncover concealed fraud groups [7], meanwhile indicating Return Rate Percentage which delivers straightforward numerical evaluations of exposure [9]. According to recent findings, an agreement has been reached: artificial intelligence excels in detecting trends - yet proves most valuable while guiding decisions

made by individual professionals. Systems project greatest impact, compared to model described here, transform intricate information into understandable reports called conclusion aimed at key users [10].

3 Methodology

Despite common approaches used, this tool finds fraud lent consumers using layered module in deep neural networks. It originates from flask lightweight module; the programming setup includes live data collected along with a detailed behavior tracking system across user sessions. One thing worth noting is that decisions don't emerge from any isolated events but from patterns that are drawn over a long period of time which are also guided by methods that are referenced in [3] and refined using comparison strategies based on research paper [5]. Instead of solely relying on hit-or-miss judgements, this approach measures the trustworthiness along with shifting scales. Central to its function, one component calculates precise returns of potential returns that are fraudlent while supporting an interactive guide at the interface level, feature shaped partially in [11]. Rare in similar tools available, this balances the detecting mechanisms along with usability considerations that does not favor one over the other and gives a fair output. This method uses two layers so the system can track any fraudlent or deliberate return inconsistencies [2], yet it delivers practical tools like pricing that checks for an item across all platforms and provides a list that can be referred to buy item at a cheap rate. Since the main task is completed by XGBoost and Neural Networks, the detection of irregular fraudulent behaviors is made possible; at the same time a rule leaded NLP mechanism converts all the basic results into clear written summaries that can be presented in a tabulated format. In conclusion, transparency is highly maintained, meaning explanations stay accessible to admin staff.

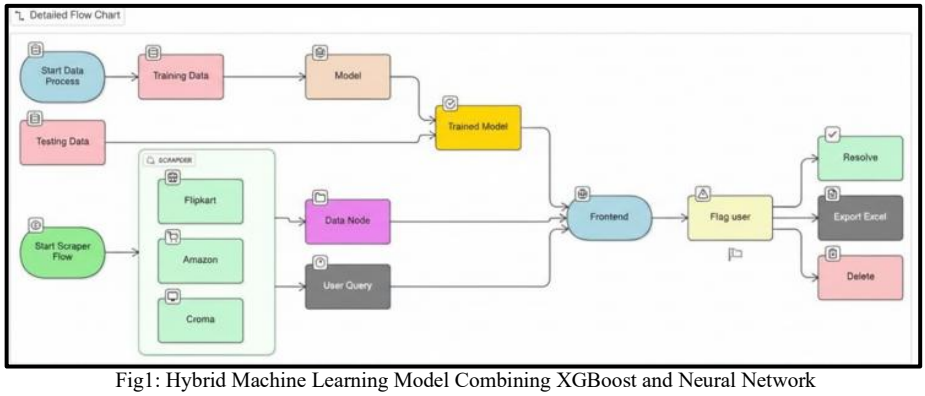

Fig1: Hybrid Machine Learning Model Combining XGBoost and Neural Network

Fig 1 illustrates how Fraud Lens operates. Starting simultaneously, one part of the project handles past records to refine all the gathered prediction. Meanwhile, other accumulates current marketplace details from sites such as Amazon and Flipkart with little to no delay. These paths converge in the data node which occurs prior to presentation on an interface. When a request is received, evaluation starts right away! Basic questions occur

in the system such as should irregularities appear, activate alert under the flag user procedures. Admins then are presented with choices such as export the conclusions in a spreadsheet in xlsx format, address concerns directly, or remove entries all together.

3.1 Core System Architecture and Technologies

In the center lies a unified structure that is formed by very powerful tools from the open community. This framework combines components that are designed for speed by collaborative development efforts. Built without any restrictions it relies on see through system which work together seamlessly. Performance emerges carefully by linked elements that shares a common standard. Each part is connected to ensure reliability across varied environments. Foundational strength comes from repetitive interactions among independent yet interdependent node modules.

Mainly the core system structure relies heavily on Flask, which redirects RESTful API calls that links machine learning components. From session tracking to previous transactions, data organization runs in Flask-SQL Alchemy. A PostgreSQL module stores information which is arranged with a lot of precision. Integration unfolds in each step which ensures consistency across all operations. A core structure relies on Flask, directing RESTful API calls while linking machine learning components. From session tracking to past transactions, data organization runs through Flask-SQL Alchemy. A PostgreSQL system holds the information, arranged with precision. Integration unfolds behind each interaction, ensuring consistency across operations.

A machine learning engine operates via two components that works simultaneously. First half applies XGBoost to manage boosted decision trees. Meanwhile the other half uses a Sequential Neural Network - developed with keras and TensorFlow - for capturing complex feature patterns. This structure allows the system to find different aspects of prediction tasks.

Periodically, live pricings of items are collected by automated methods. One tool processes the site's layout by interpreting a raw markup directly. At the same time, other handles communication across multiple other networks and requests efficiently. Together this enables an efficient and continuous monitoring without manual involvement. These system components are operated independently but are aligned closely during execution cycles. This integration supports access to be updated Figs online. Each part performs under a variable condition while maintaining the overall accuracy of the system. The efforts rely on structured navigation which helps in efficient navigation throughout the system rather than the presence of visual cues throughout. Despite separate roles, synchronization remains very stable even at a scale. This method is adaptive quickly when sources have modified their output format unexpectedly. Throughout operation, minimum overhead is preserved systematically. Such design allows very efficient as well as reliable extraction despite updates that occur elsewhere.

By keeping the security in mind, consumer identifiers appear only as hashed values. Though relations are important, the structural links Customer TO Order History and Referral Flags without any gaps. Consistency arises from stability - data ties remain intact by design.

3.2 Data Acquisition and Knowledge Base Creation

In this step, there are two types of information of which one enters through a dedicated channel, while another passes through a separate path entirely. Their entry methods differ in structure and timing, still they are both handled by the same framework, in contrast, it only triggers when the second activates. They converge later within the core mechanism yet every route maintains its own rules.

Internal Fraud Information Collection Transactions

This dataset is derived from the purchase records of various e-commerce sites. The return pattern of the product along with the buyers' information is present in it. This knowledge-base is arranged systematically. It is essential to note that this dataset is a not live and it is to several seasons. The entries are neither manually entered nor registered. After the checks from the verification processes that ensure temporal consistency, the information enters into the structure. Each record represents the member of an ensemble derived as a result of past observations. It could be the result of automated logging from a sequence of purchase, review cycles. After each cycle, this size continues to grow.

A series of steps must be taken before the analysis begins on the raw inputs. It normalizes the numerical values within the pipeline. Encoded Representation Technique Used To Convert Categorical Fields. During processing, certain features are extracted, for example, frequency of return and average cost of item. The machine learning structure receives these outputs for further procedure.

Because fraudulent activity happens at much lower rates than valid activity, the synthetic data generation algorithm manipulates these proportions in each class. Thus, assumptions collected under one category will not influence the interpretation under a different category. This means we attain equilibrium by replication that is of structural kind and not of natural kind.

Market Intelligence Base Price Corpus

Live pricing data obtained from Amazon, eBay, Flipkart, and Myntra. A dedicated scraping process links each product SKU to many other platforms. The price tag, currency, and availability status are extracted using BS4. The information is stored temporarily through redis. Instant access for price comparison functionality has been activated. If you already scraped an item, wait at least 30 minutes after the last fetch.

3.3 Module-Specific Implementation

Hybrid Refund Abuse Detection

The XGBoost model is first used on structured data, such as stated rates of returned orders. Subsequently, a neural architecture analyses the sequence of timed interactions. Instead of relying on just one technique, the results from these two stages combine according to weighted voting principles. Notifications are automatically sent to human evaluators once the likelihood scores exceed thresholds. Through phases of joint analysis of distinct strength of the design arise.

Risk Quantification Module the RRP Engine

The customerID applies mathematical logic to calculate the percentage return rate. This alert gets activated when the (RRP) exceeds 40% value. Because of the threshold, the website identifies frequent returners without triggering a serious alarm.

$$RRP = \left(\frac{\sum Returned_Items}{\sum Total_Orders}\right) \times 100$$

Cross Platform Price Comparison Tool

This function helps with access and tracking of actions. Through FuzzyWuzzy comparison, matches are formed between items listed. The person seems to get a sorted summary of favorable options. Internal records capture how pricing influences decisions. This provides a more precise view of individual decision patterns over time. Although things appear consistent, subtle shifts in response can have profound long-term adjustments behind the scenes.

4 Results

4.1 Navigation Hub

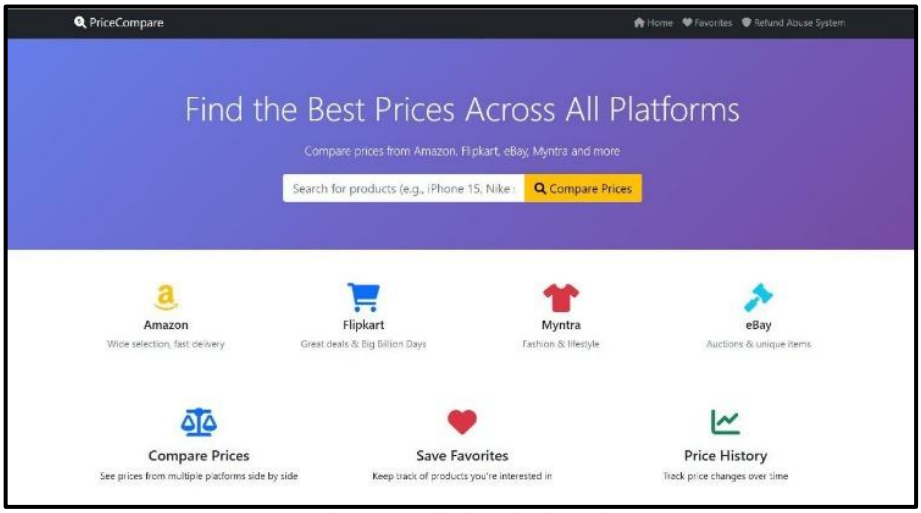

Fig 2 : Centralized Navigation Hub

A master screen at the top gives you options that spread out logically from the master screen. Using the tools feels intuitive, as everything exists inside discrete blocks that have padding inside and between them. One direction allows for price comparisons among different stores, while the other provides controls to identify refund abuse. There are other places of personal importance like saved favorites that are tucked away out of sight. There is a smooth and continuous transition from aid shopping to safety auditing.

4.2 Administrative Command Centre

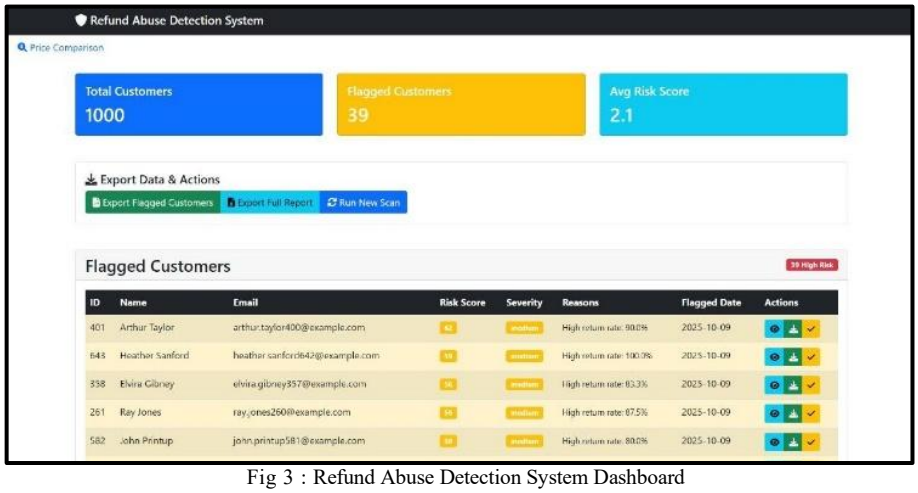

Fig 3 : Refund Abuse Detection System Dashboard

The far-right section of the admin screen portrays a zoomed-out perspective depicting the potential threat to the system. Beginning with the numbers we see overall stats that show total users at one thousand, thirty-nine flagged cases, with overall average danger flag at two point one. Below this, there is a Flagged Customers panel set up to run on RRP rules that reveals a lot more detail. We can see return rate from ½ to 87.8%. This causes an auto flag for warning but staff are allowed to pull up records for their own investigations if they feel that it is necessary.

4.3 Consumer Engagement Portal

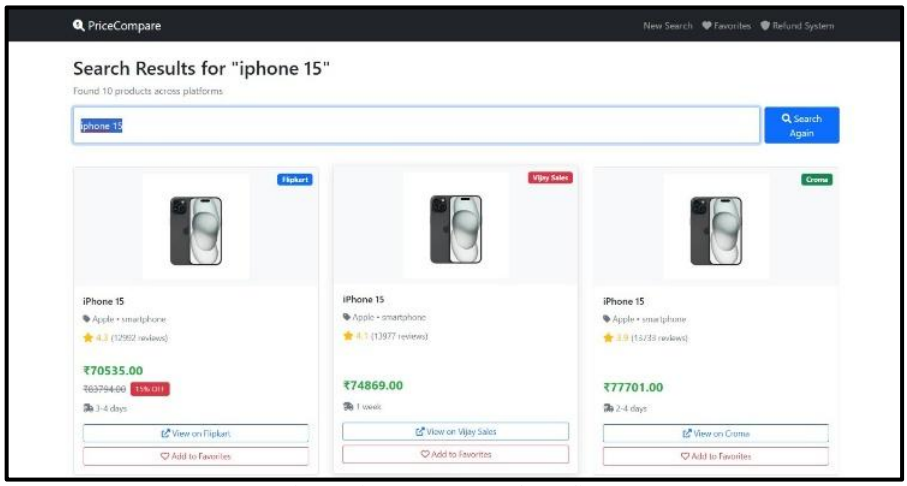

Fig 4 : Price Compare Search Landing Page

The far-right section of the admin screen portrays a zoomed-out perspective depicting the potential threat to the system. Beginning with the numbers we see overall stats that

show total users at one thousand, thirty-nine flagged cases, with overall average danger flag at two point one. Below this, there is a Flagged Customers panel set up to run on RRP rules that reveals a lot more detail. We can see return rate from ½ to 87.8%. This causes an auto flag for warning but staff are allowed to pull up records for their own investigations if they feel that it is necessary.

5 Conclusion

Every security system has to have alarms. Instead of going behind fraud after it has been occurred, keeping watch over how user act reveals more vital information. A mixture of smart models - one which has accurate pattern recognition, another tuned for subtle shifts - works in the background. This combination of smart and subtle mechanism spots fraud lent returns using a clear number: how often someone is availing the money return policy. At the same time shoppers get help comparing prices across the internet. One part feeds info to another part while it oversees every move the person makes. Profit stays protected when model is trained enough. Trust between consumer and seller grows when friction fades.

Computer vision may be used in the future to verify the appearance of returned goods, identifying anomalies such as used clothing returned as brand-new. Smart summaries created by language models might provide investigators with more detailed images of cunning conduct rather than only forms. Live scanning of customer support interactions might reveal genuine annoyance as opposed to rule-gaming. The goal of these changes is to improve detection while maintaining honest results. When algorithms are able to read between the lines, deeper hints become apparent. Today, most tools overlook the minutiae that fraud lies in.

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